Evaluation ensures recommendations are not just visually appealing, but also statistically meaningful.

Streamlit deployment on Linux

A recommendation engine becomes truly useful only when users can interact with it. I built a lightweight frontend using Streamlit, an open source Python framework. Streamlit runs smoothly on Linux, needs minimal code, has a real-time interactive UI, and is easily deployable on servers.

The user experience is:

- Enter a book title.
- Choose model type.
- Get instant recommendations.

Streamlit turns ML projects into practical applications.

Deployment architecture

This project can be deployed on open source or cloud infrastructure. Deployment options include:

- Linux VPS server
- AWS EC2 instance
- Elastic Beanstalk
- Docker container

The architecture pipeline is: *Dataset → Cleaning → NLP → Clustering → Recommendation Models → Streamlit UI → Deployment.*

This modular design ensures scalability and maintainability.

You've learnt how open source machine learning tools can help to build a hybrid recommendation engine. Such recommendation systems are not only valuable in audiobooks but can also be extended to:

- Movie platforms
- News aggregators
- E-commerce catalogues
- Academic paper recommendations

The best part is, with open source tools, developers have the complete freedom to build, customise, and deploy intelligent systems without proprietary dependencies. END

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The author is a software engineer with hands-on experience in full-stack and machine learning projects. He is passionate about open source and cloud-based AI systems.